Leveraging Web Scraping and AI Agents For Scam Detection on Social Media

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• QUESTIONS

1. **How can AI-Agents and web scraping techniques be combined to effectively track evolving scam activity on social media and communication platforms?**
2. **Does a Closed Loop Knowledge Sharing Multi Agent architecture (CLMA) outperform an Open Loop Feed forward pipeline (FFP) architecture for real time scam detection?**

• BACKGROUND

TOTAL MONETARY LOSS	LEADING CONTACT METHOD	PERCENTAGE OF RECOVERED LOSS	STAGNANT SOLUTIONS TO EVOLVING TACTICS
$1.03 Trillion USD in 2024	Social Media at 36%	Only 4% of all Funds Lost where Recovered	Current Solutions are Reactionary

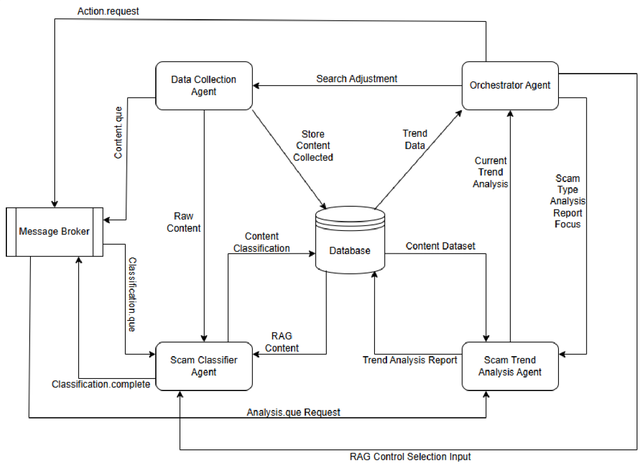
Social Media Platforms currently employ statis rule-based moderation enforced with the use of AI and reporting metrics to combat scam activity

Scammers continuously evolve their methods to avoid detection from platforms and learn from repeated failures.

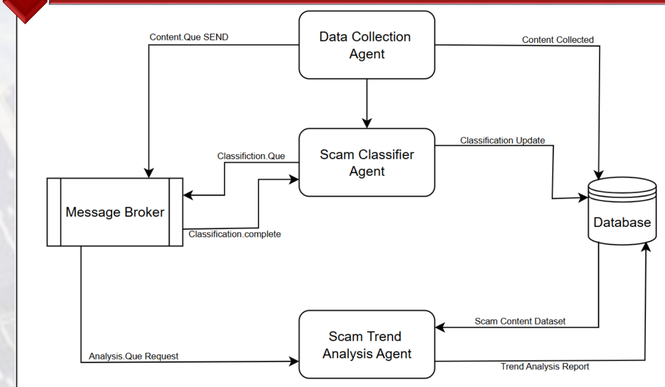
This presents unique software engineering challenges in developing an adaptative solution that can operate in real time.

Organizations across multiple industries facing the financial damages due to social media scams strongly emphasize the need for a solution.

• CLOSED LOOP ARCHITECTURE



• OPEN LOOP ARCHITECTURE



• SYSTEM OPERATIONS

DATA COLLECTION AGENT	SCAM CLASSIFIER AGENT	SCAM TREND ANALYSIS AGENT	ORCHESTRATOR AGENT
Web Crawling and Scraping	Data Processing and Classification	Scam Trend Analysis and Quantitative Measures	Evaluation of Results and Process Adjustments

• TESTING AND RESULTS

Closed Loop Multi Agent System Results

Module / Agent	Metric	Result (Target)	Pass?	Notes
Data Collection Agent	Recall	0.92 (≥ 0.89)	Yes	Compared against baseline process results
	Discovery Success Rate	avg 25/min (≥ 10/min)	Yes	Observation → successful process
Classifier Agent	ROC–AUC	0.92 (≥ 0.84)	Yes	Ranking on test dataset
	F1@τ	0.84 (≥ 0.78)	Yes	τ fixed from config; confusion matrix verified
Trend Analysis Agent	Cluster labeling	0.98 (≥ 0.85)	Yes	Analyst-verified trend alerts
	Trend detection lag	2 cycles (≤ 3 cycles)	Yes	New pattern surfaced to analysts
Orchestrator Agent	Action lag	5 min (≤ 5 min)	Yes	Time between final results and adjustments
	Idempotency success	99.95% ($\geq 99.9\%$)	Yes	Duplicate config doesn't create duplicate outputs

Open Loop Feed Forward System Results

Module / Agent	Metric	Result (Target)	Pass?	Notes
Data Collection Agent	Recall	0.89 (≥ 0.85)	Yes	Compare to previous cycle results
	Discovery Success Rate	avg 12/min (≥ 10/min)	Yes	Observation → successful process
Classifier Agent	ROC–AUC	0.84 (≥ 0.80)	Yes	Ranking on test dataset
	F1@τ	0.78 (≥ 0.75)	Yes	τ fixed from dev; confusion matrix verified
Trend Analysis Agent	Cluster labeling	0.89 (≥ 0.85)	Yes	Analyst-verified trend alerts
	Trend detection lag	3 cycles (≤ 5 cycles)	Yes	New pattern surfaced to analysts

- **Data Collection** produced a continuous 4.5% increase in the number of new items searched every hour in the Closed Loop Multi Agent System.
- **The Classification Agent** produced a 92% accuracy score when implemented with selective RAG exemplar (Same dataset used for both architecture tests).
- **Scam Trend Analysis Agent** progressively improved categorizing new records if clusters maintained form over time.
- **The Orchestrator Agents** influence produced an F1 score 2-7% higher per call every 3 iterations in the real world testing procedure.